Lab 10

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Lab 10

prelab
DG
4/20

10.1
(a) picture in submission

(b) verified the LED lights up

(c) imported and set up!

(d) By default, it uses a sine wave, which we set to 1kHz, 2V

(e) LED flashes!

10.2
(a) oscilloscope_run takes in an ADSHardware object, a sample time, and a sampling rate

(b) The x axis should be time, but it is instead plotting the sample number!

(c) take data["x"] *= MS_conversion / sample_rate

(d) corrected plot in full lab submission

10.3
(a) done!

VS 4/15

(b) abs(fft(data["y"], norm="forward")) and fftfreq(dt, data["x"]) work well

(c) re-filtered ✓

(d) plot in full lab submission

(e) ↳ norm factor should be correct I think

(f) xlims set - plot in final submission

(g) done

10.4

(a) built! photo in lab submission

(b) DC offset set

(c) oscilloscope set up, plots in full lab submission

(d) data exported

10.5

(a) ✓

(b) done - collected data with DC offset, the modulated signal looks fantastic!

(c) filled in - code in final lab submission

(d) collected data with $V_{sdrs} = 10$ Hz

(e) used oscilloscope in checkoff

(f) plots in final lab submission - our modulated signal is worse than the raw DC signal

frequency: 1.33 Hz, 80 bpm

10.6

(a) the lockin function needs to turn the wavegen on and off repeatedly as well as collect the signal and noise in different bins, so it takes control of use_wavegen and oscilloscope_run

blanks filled in

(b) plot in final lab submission

(c) lockin bpm: 78.57 bpm, bad SNR after fft

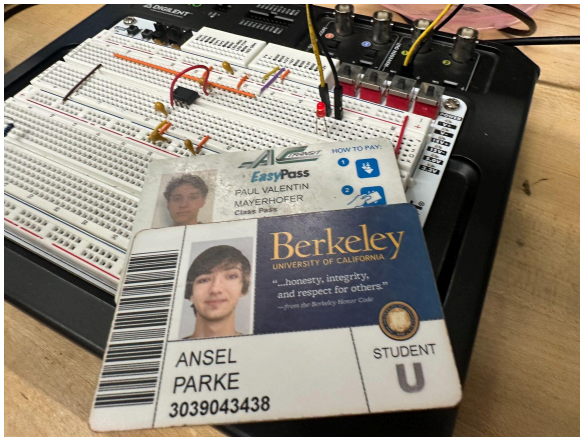
radio demod bpm: 78.57 bpm, best SNR after fft

DC baseband bpm: 64.28 bpm, second best SNR

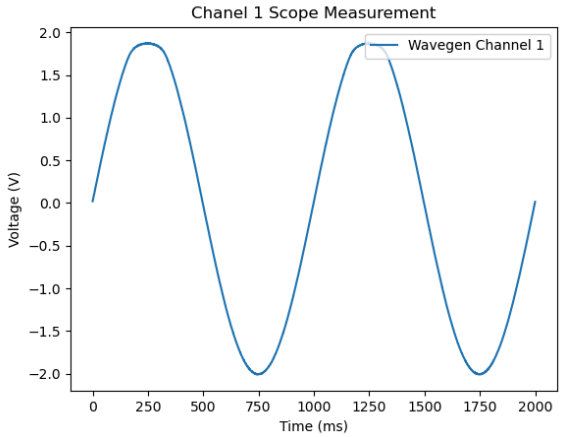
VS

OW/LS

10.1



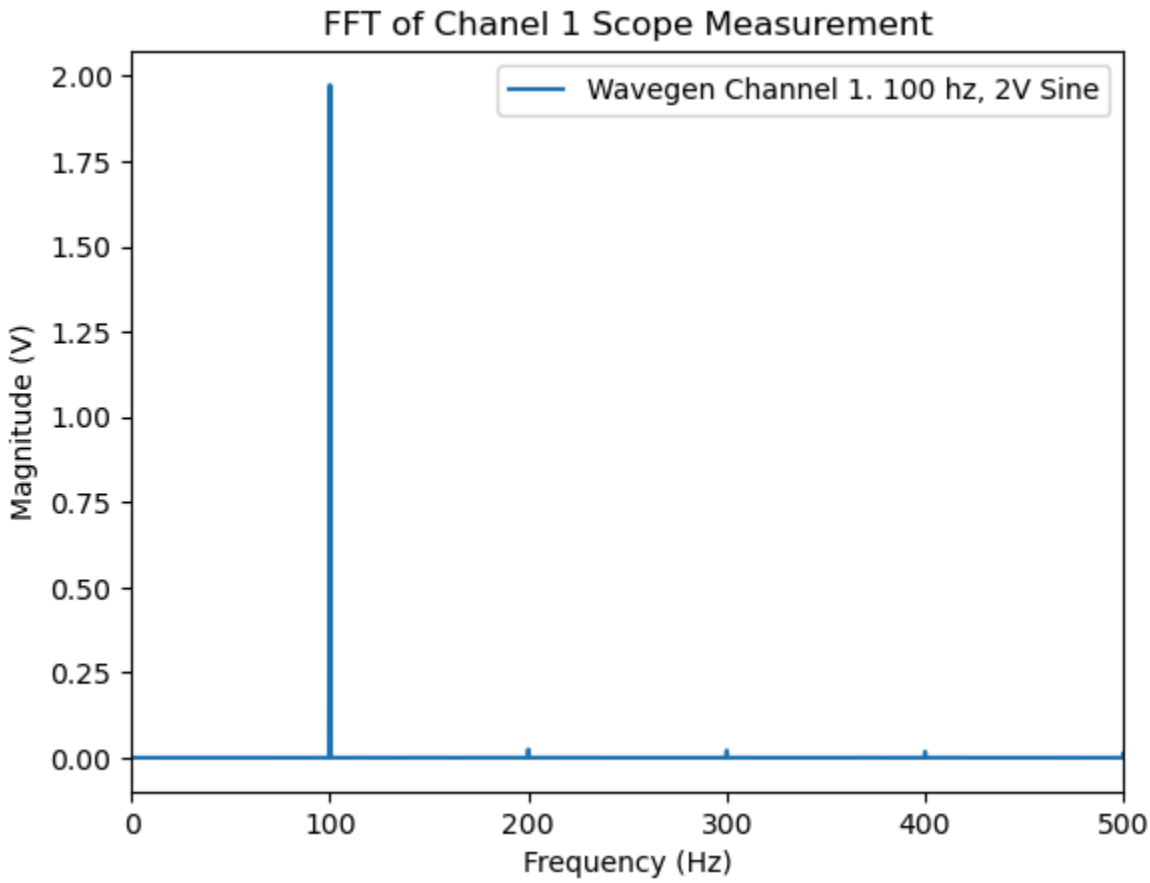
10.2



raw output of oscilloscope connected to wavegen

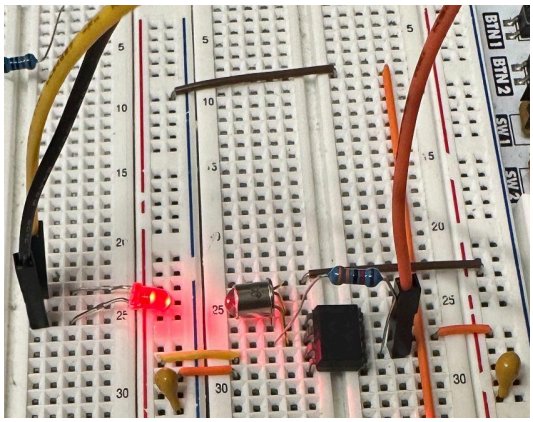
10.3

d-g)



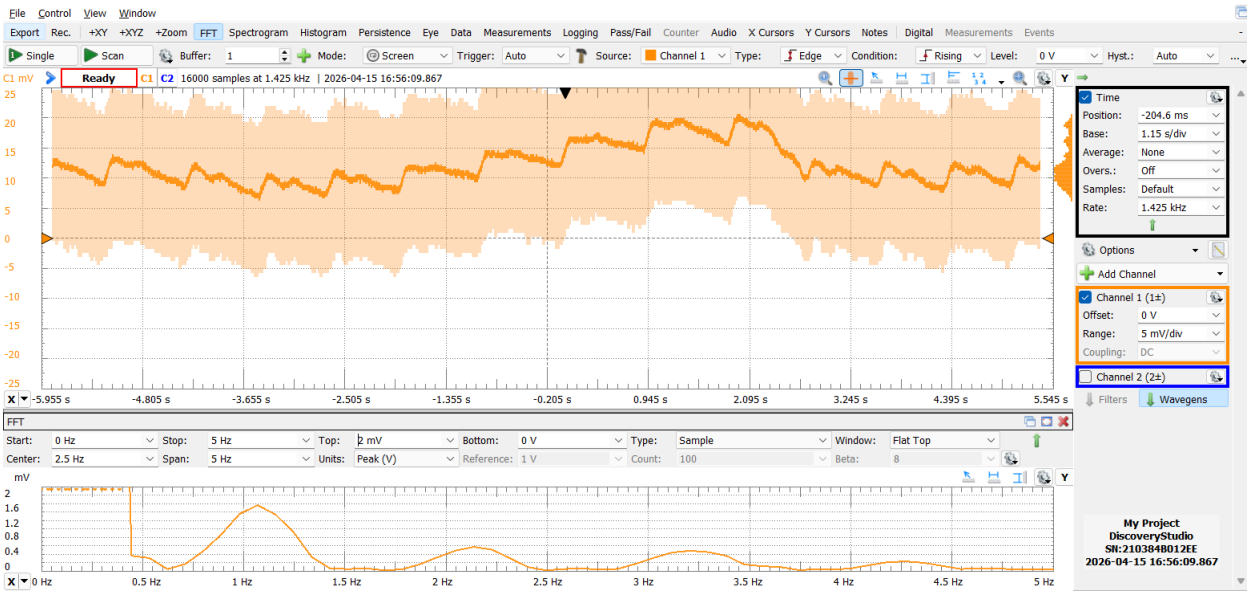
2k sample rate over 2 seconds at 100mV and 100Hz

10.4

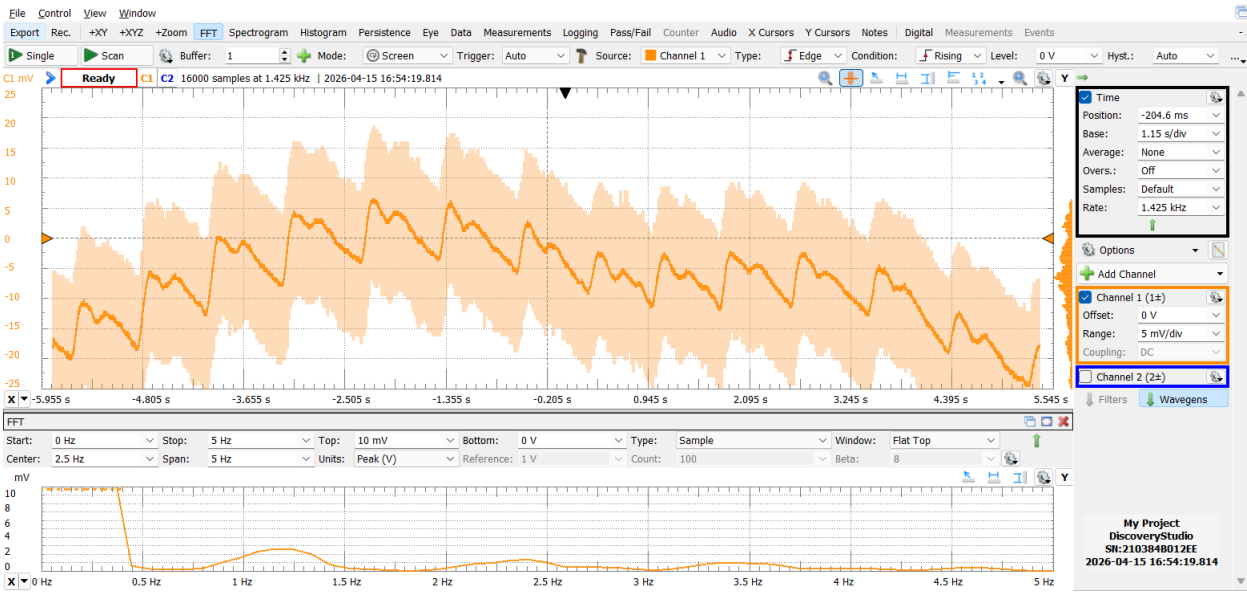


b) Setup scope=2x16k, zoomed in around 1 Hz on fft

i) Paul's Heartbeat

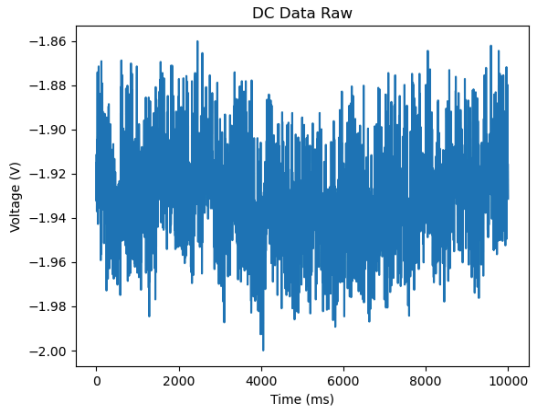


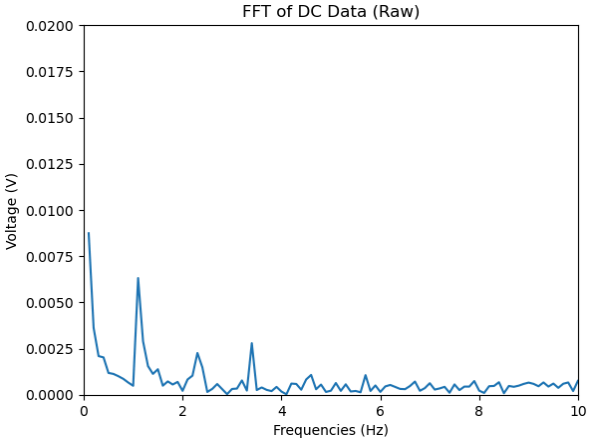
ii) Ansel's Heartbeat



10.5

b)

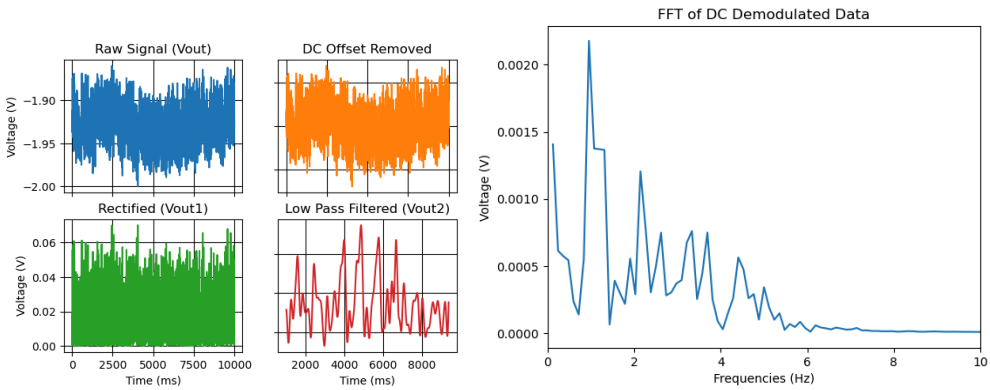




i) `First peak is: 1.10 Hz or 66.0 Bpm`

DC Baseband data with FFT

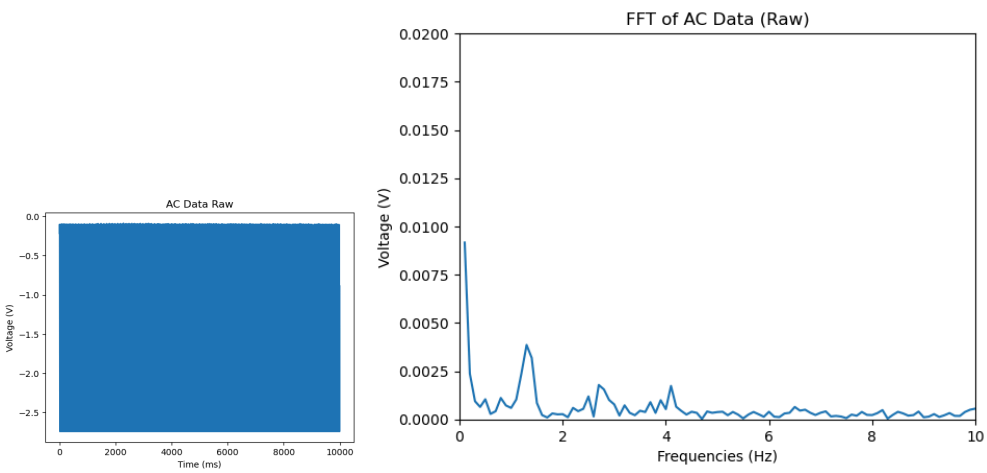
C)



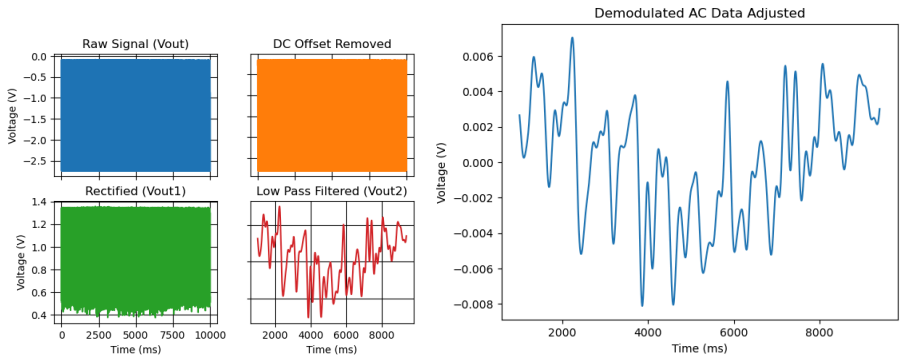
Demodulated data from DC signal, with better fourier peak
Gives 1.31 +/- 0.1 Hz or 78.6 +/- 6 BPM.

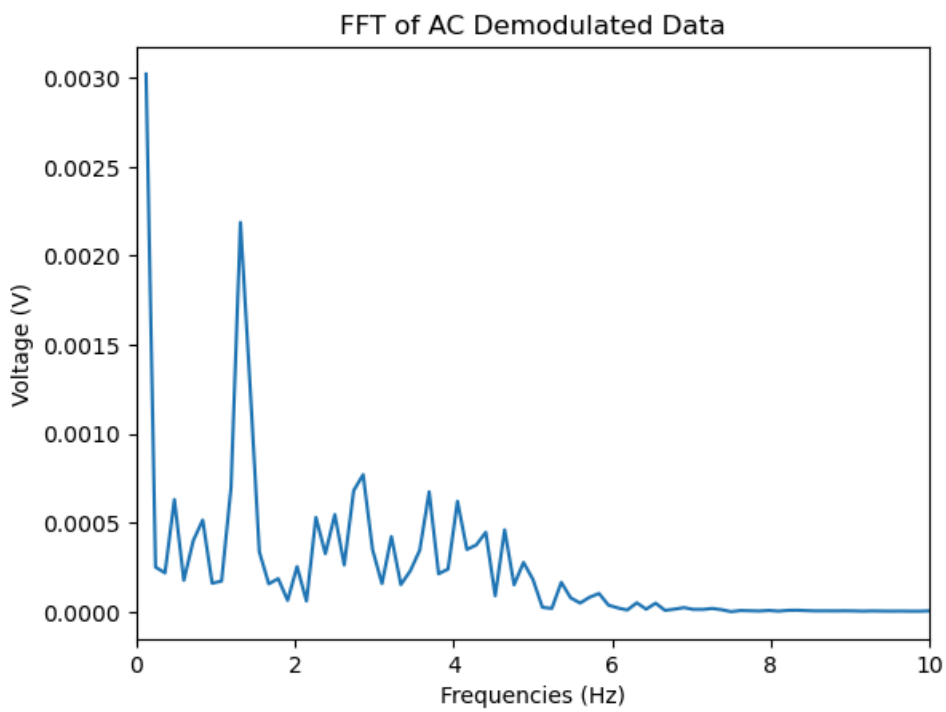
d) Running the main code with the 50 hz 2.75 V sine carrier wave we got:

Before Radio Demodulation:



After Radio Demodulation (we set our 3db to 5 Hz):

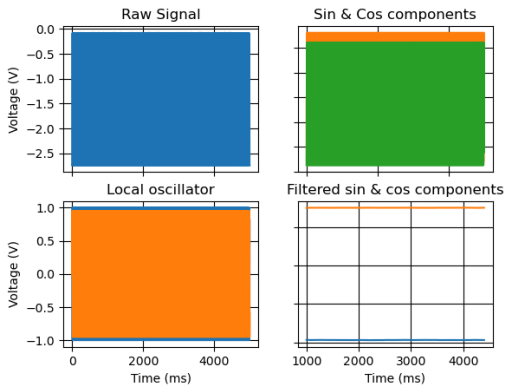




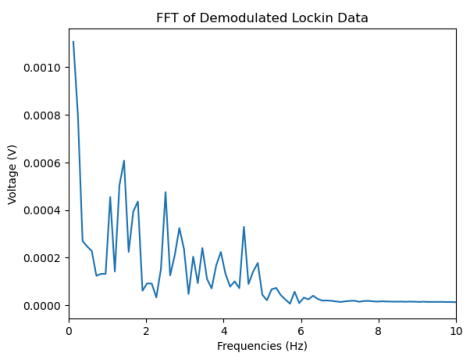
We see the heartbeat pretty clearly at 1.31 Hz or 78.6 BPM and its harmonics! Our error is 0.1 hz or 6 BPM due to the 10 second sample size limiting the FFT decomp.

10.6

b) We used a modulation frequency of 50 Hz and a 3dB frequency of 7Hz to attenuate the modulation.

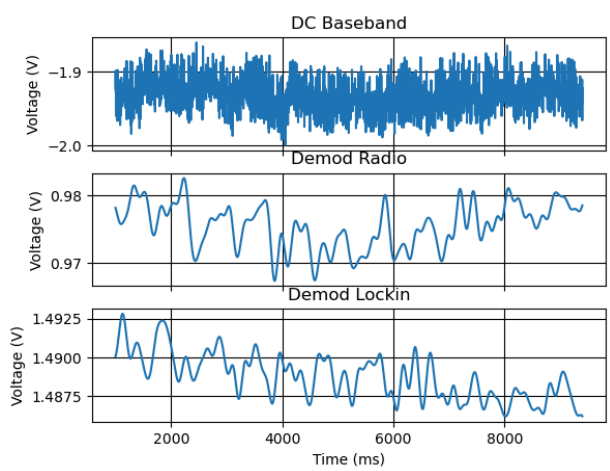


Intermediary data from demodulate_lockin

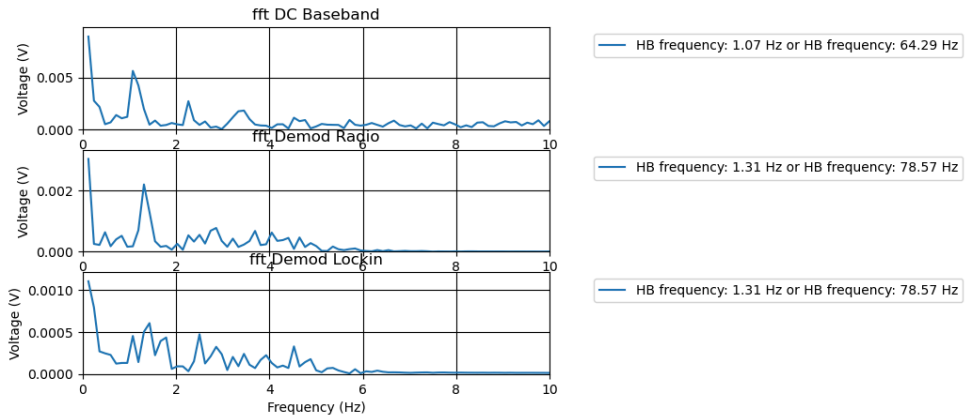


FFT of post-demodulate_lockin data

c)



DC non-demodulated data, AC demodulated data with simple demodulation and lock-in demodulation respectively, from top to bottom



Comparison of FFTs for each method